Exercise Sheet 22

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$ and

$$k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $\mathbf{e}_1 = (1,0)^{\mathsf{T}}$, $u_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}}$ and $e_n(\theta) = e^{in\theta}$. The inner product is given by

$$\langle \varphi, \psi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \overline{\varphi}\, \psi \, d\theta.$$

It is antilinear in the first slot. Each exercise recalls any further data it uses from Lecture 22.

Exercise 22.1. This exercise computes the Iwasawa decomposition $G = KAN$.

(a) Show that every $g \in G$ can be written uniquely as $g = k_\theta a_t n_x$ with $\theta \in \mathbb{R}/2\pi\mathbb{Z}$ and $t, x \in \mathbb{R}$. Show also that $g\mathbf{e}_1 = e^{t} u_\theta$.

(b) Find θ, t and x for

$$g = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix},$$

and check the product.

(c) Let $f : G \to \mathbb{C}$ satisfy $f(g\, m a_s n_y) = \sigma_\varepsilon(m) e^{-(\nu+1)s} f(g)$ for $m = \pm I$, where $\sigma_\varepsilon(\pm I) = (\pm 1)^{\varepsilon}$. Put $\varphi(\theta) = f(k_\theta)$. Express $f(g)$, for the matrix g of (b), through φ.

Exercise 22.2. Let P be the group of upper triangular matrices in G, so $P = \{ m a_t n_x : m = \pm I \}$. Its tangent algebra has basis $H = \mathrm{diag}(1,-1)$ and $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Use the modular convention $\int f(xg)\,dx = \Delta(g)^{-1} \int f(x)\,dx$, for which $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$.

(a) Compute the matrices of $\mathrm{Ad}(a_t)$, $\mathrm{Ad}(n_x)$ and $\mathrm{Ad}(-I)$ in the basis (H, E). Deduce $\Delta_P(m a_t n_x) = e^{-2t}$.

(b) Let Aff be the group of pairs (a, b) with $a > 0$ and law $(a,b)(a',b') = (aa', b + ab')$. Its modular function is $\Delta(a,b) = 1/a$. Show that $\psi(m a_t n_x) = (e^{2t}, e^{2t} x)$ is a surjective homomorphism $P \to \mathrm{Aff}$ with kernel $\{\pm I\}$, and that $\Delta_P = \Delta_{\mathrm{Aff}} \circ \psi$.

(c) Let $\chi_{\varepsilon,\nu}(m a_t n_x) = \sigma_\varepsilon(m) e^{\nu t}$. Show that $\Delta_P(p)^{1/2} \chi_{\varepsilon,\nu}(p)^{-1} = \sigma_\varepsilon(m) e^{-(\nu+1)t}$ for $p = m a_t n_x$. Show that this has absolute value squared $\Delta_P(p)$ when $\nu \in i\mathbb{R}$.

Exercise 22.3. For $h \in G$ define τ_h on the circle by $h u_\theta = |h u_\theta|\, u_{\tau_h(\theta)}$. You may use the fact, proved in Lecture 22, that each τ_h is smooth. Fix $x \in \mathbb{R}$ and put $h = n_x^{-1} = n_{-x}$.

(a) Compute $h u_\theta$ and $r(\theta) = |h u_\theta|$. Show that $\cot \tau_h(\theta) = \cot\theta - x$ when $\sin\theta \neq 0$.

(b) Verify $\tau_h'(\theta) = r(\theta)^{-2}$ by differentiating (a). Explain why the identity also holds where $\sin\theta = 0$.

(c) For $A, B > 0$, prove that

$$\int_0^{2\pi} \frac{d\theta}{A\cos^2\theta + B\sin^2\theta} = \frac{2\pi}{\sqrt{AB}}.$$

For every $\nu \in \mathbb{C}$ the circle model gives $(\Pi_{0,\nu}(n_x) e_0)(\theta) = r(\theta)^{-(\nu+1)}$. For $\nu \in i\mathbb{R}$, use this to show directly that $\| \Pi_{0,\nu}(n_x) e_0 \| = 1$.

Exercise 22.4. In $\mathfrak{sl}(2,\mathbb{R})$ let

$$W = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and $X_\pm = H \pm iQ$. On $V_{\varepsilon,\nu} = \operatorname{span}\{e_n : n \equiv \varepsilon \pmod 2\}$, the ladder formulas of Lecture 22 read

$$d\Pi(W)e_n = -in\, e_n, \qquad d\Pi(X_+)e_n = (\nu + 1 + n)\, e_{n+2}, \qquad d\Pi(X_-)e_n = (\nu + 1 - n)\, e_{n-2}.$$

(a) Compute $[W,H]$, $[W,Q]$ and $[H,Q]$. Deduce $[W,X_\pm] = \mp 2iX_\pm$ and $[X_+,X_-] = 4iW$.

(b) Check on each e_n the three identities

$$\begin{aligned} d\Pi(W)d\Pi(X_+) - d\Pi(X_+)d\Pi(W) &= -2i\, d\Pi(X_+), \\ d\Pi(W)d\Pi(X_-) - d\Pi(X_-)d\Pi(W) &= 2i\, d\Pi(X_-), \\ d\Pi(X_+)d\Pi(X_-) - d\Pi(X_-)d\Pi(X_+) &= 4i\, d\Pi(W). \end{aligned}$$

(c) Let $\Omega = H^2 + Q^2 - W^2$ in $U(\mathfrak{g}_{\mathbb{C}})$. Show $X_+X_- = H^2 + Q^2 + 2iW$. Deduce that Ω acts on $V_{\varepsilon,\nu}$ by $\nu^2 - 1$. Which other parameter gives the same scalar?

Exercise 22.5. Take $\varepsilon = 0$. By the submodule criterion of Lecture 22, a subspace of $V_{0,\nu}$ is a $(\mathfrak{g},K)$-submodule exactly when it is $\operatorname{span}\{e_n : n \in S\}$ for a set S of even integers such that the following two conditions hold. If $n \in S$ and $\nu + 1 + n \neq 0$, then $n+2 \in S$. If $n \in S$ and $\nu + 1 - n \neq 0$, then $n - 2 \in S$.

(a) List all submodules of $V_{0,-1}$. Identify one on which $\mathfrak{g}$ acts by 0 and K trivially.

(b) List all submodules of $V_{0,1}$.

(c) Let $\nu \in \mathbb{C}$. For $g \in G$, write $g^{-1}u_\theta = |g^{-1}u_\theta|\, u_{\tau(\theta)}$. In the circle model,

$$(\Pi_{0,\nu}(g)\varphi)(\theta) = |g^{-1}u_\theta|^{-(\nu+1)}\, \varphi\big(\tau(\theta)\big).$$

Let φ, ψ be continuous functions with $\varphi(\theta+\pi) = \varphi(\theta)$ and $\psi(\theta+\pi) = \psi(\theta)$. The pairing identity of Lecture 22 states

$$\langle \Pi_{0,-\bar\nu}(g)\varphi, \Pi_{0,\nu}(g)\psi\rangle = \langle \varphi, \psi\rangle.$$

Show that $\ell(\psi) = \langle e_0, \psi\rangle$ satisfies $\ell(\Pi_{0,1}(g)\psi) = \ell(\psi)$. Compare $\ker \ell \cap V_{0,1}$ with (b).

Exercise 22.6. Use the ladder formulas of Exercise 22.4 on $V_{\varepsilon,\nu}$, with the inner product $\langle e_m, e_n\rangle = \delta_{mn}$. An operator D on $V_{\varepsilon,\nu}$ is *skew-symmetric* if

$$\langle \varphi, D\psi\rangle = -\langle D\varphi, \psi\rangle$$

for all $\varphi, \psi \in V_{\varepsilon,\nu}$.

(a) Express $d\Pi(H)$ and $d\Pi(Q)$ through $d\Pi(X_\pm)$. Show that $d\Pi(H)$ is skew-symmetric if and only if $\nu \in i\mathbb{R}$.

(b) Show that $d\Pi(W)$ is always skew-symmetric, and that $d\Pi(Q)$ is skew-symmetric if and only if $\nu \in i\mathbb{R}$.

(c) For $\varepsilon = 0$ and $\nu = 1$, compute $d\Pi(H)e_0$ and $d\Pi(H)e_2$. Exhibit the failure of skew-symmetry.

Exercise 22.7 (Optional extension). Take $\varepsilon = 0$ and real ν with $\nu \neq \pm 1$. Use the ladder formulas above. Look for a new inner product on $V_{0,\nu}$ with $\langle e_m, e_n \rangle_c = c_n \delta_{mn}$, where $c_n > 0$ and $c_0 = 1$, such that $d\Pi(W)$, $d\Pi(H)$ and $d\Pi(Q)$ are skew-symmetric for $\langle \cdot, \cdot \rangle_c$.

(a) Show that the condition is $c_{n+2}(n + 1 + \nu) = c_n(n + 1 - \nu)$ for all even n.

(b) For $|\nu| < 1$, show that a solution exists with $c_{-n} = c_n$, and give c_n for $n > 0$. Compute c_2 and c_4 for $\nu = \frac{1}{2}$.

(c) For $|\nu| > 1$, show that no positive solution exists.